

STRIPS represent a collection of zero-coupon bonds that are sold at a discount of their face value. The return they provide is reflected as the growth rate on price paid, and the face value is received at the maturity date. The price of a zero-coupon bond slowly gravitates toward its face value as the maturity date approaches.

The Treasury facilitates the creation of STRIPS by “stripping” the coupon payments and face value payment from Treasury bonds and selling them as separate securities. Each coupon payment becomes a standalone zero-coupon security providing the one coupon payment at its specified date (Coupon STRIPS). The return of the face value at the maturity date also becomes a zero-coupon security (Principal STRIPS).

Bonds with more distant maturity dates typically offer higher interest rates than bonds with earlier maturity dates. While that is not always the case, Exhibit 7.3 shows us it was true in January 2017, as thirty-year STRIPS were yielding 3.1 percent. Longer-term bonds experience bigger price fluctuations as interest rates change, which explains why the 2046 yield is noticeably higher than earlier maturities. When interest rates increase, the price of existing bonds on the secondary market falls in order to calibrate the yield investors will receive from owning existing bonds with the yields being offered by newly issued bonds at the higher interest rates. Bonds that mature sooner are less exposed to this price risk. Thus, shorter-term US Treasury securities are generally considered to be among the lowest-risk investment assets when annualized volatility represents the measure of risk, which means they tend to be offered at a lower yield. Higher yields accompany longer-term bonds, as investors need an added incentive to accept the higher price risks.

The shape of the yield curve is molded by two theories. First, expectations theory suggests that the shape of the yield curve should be reflected by beliefs about future short-term interest rates. Investing in bonds over ten years can be done in two ways:

1. Buy a ten-year bond, or
2. Buy a one-year bond and then reinvest in a new one-year bond after one year, continuing with a succession of ten one-year bonds.